Standard model, published construction

Combined factor-model portfolio test

Practitioner factor-model runner

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Abstract

Model. Standard model, published construction, one long-short book from 4 sleeves (VAL_rebuilt, MOM_rebuilt, BAB_rebuilt, QMJ_rebuilt). 5857 symbols (full screened universe), 2008-11-01 to 2026-07-01 (213 months).

Earned. 3.32% annual arithmetic return (3.18% compounded), 6.02% volatility, Sharpe 0.55, HAC t=2.16.

Survives? vs VTI market benchmark: alpha 5.88%, t=4.39. vs VAL+MOM+BAB+QMJ: alpha 0.48%, t=0.47. Conventional bar is $|t| \geq 2$.

Best swap. (Standard model, VAL_rebuilt for VAL) improves: Sharpe 0.86 against 0.50 (+0.35). **Verdict.** Not distinct: it clears the benchmark, but standard factors explain it.

Exploratory model diagnostic from a single configured run, not evidence from an untouched confirmatory sample.

1 Research objective

Test whether the configured model earns a useful next-month return and whether the benchmark or standard factors explain it. Signals are observed at month-end; returns are next-calendar-month simple returns.

Construction. Normalize within month, combine components within sleeves, then combine sleeves at the stock level using VAL rebuilt 25%, MOM rebuilt 25%, BAB rebuilt 25%, QMJ rebuilt 25%. Construct one portfolio; apply beta neutrality and costs once. Sleeve returns are not averaged.

2 Configuration and sample

Factor	Composite signals	Signal transform	Composite transform	Winsor	Min inputs	Mean inputs
VAL rebuilt	book_to_market	rank	rank	1%-99%	1	1.0
MOM rebuilt	momentum_21_252	rank	rank	1%-99%	1	1.0
BAB rebuilt	beta_shrunk_252v_1260c_VTI	rank	rank	1%-99%	1	1.0
QMJ rebuilt	gpoa, roe, roa, gross_margin, cfoa, accruals, growth_gpoa_5y, growth_roe_5y, growth_roa_5y, growth_cfoa_5y, growth_gmar_5y, growth_acc_5y, debt_to_assets, roe_volatility, altman_z, ivol_21d	rank	rank	1%-99%	10	14.7

Model construction	Book	Beta neutral	Bucket frac.	Cost bps	Min factors	Benchmark
signal weighted	long-short	no	0.33	0.0	4	VTI

Portfolio	Mean stocks	Minimum stocks
Combined model	2665.6	1367
VAL rebuilt	2824.0	1741
MOM rebuilt	2921.0	1863
BAB rebuilt	2630.8	1713
QMJ rebuilt	2286.4	34

Sample: 5,857 symbols, 1,049,394 stock-months, 1962-01-01–2026-08-01. Screens: price 1.0, 21-day dollar volume None, market cap None, top-cap fraction None, exclude financials False. Benchmark: VTI. Factor controls: standard_recreated_factors.csv.

3 Portfolio returns

Raw net performance on one shared window. Return and volatility use monthly arithmetic moments; t is HAC.

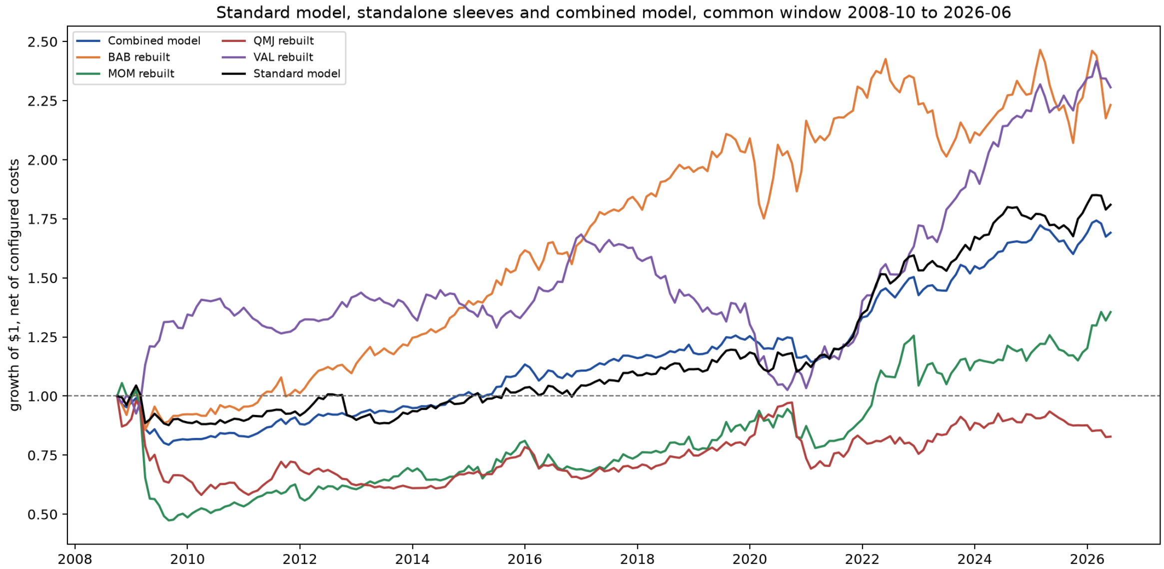

Figure 1: Growth of one dollar in each book, net of configured costs, on the common window. The path begins at one before the first return month.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	213	3.32%	3.18%	6.02%	0.55	2.16
VAL rebuilt	213	5.47%	5.11%	9.86%	0.56	2.02
MOM rebuilt	213	2.57%	1.52%	14.05%	0.18	0.66
BAB rebuilt	213	5.30%	4.92%	9.91%	0.54	2.78
QMJ rebuilt	213	-0.09%	-0.76%	11.44%	-0.01	-0.03
Standard model	212	3.62%	3.41%	7.17%	0.50	2.20

4 Benchmark fit and benchmark-orthogonalized returns

Regress each raw net return on VTI market benchmark. R2 is variance explained; alpha is the annualized intercept; *t* and F use HAC inference. This is separate from standard-factor attribution. Estimated MKT loading: -0.176. Annual raw return 3.32% = controlled alpha 5.88% + benchmark contribution -2.56%. Orthogonal means zero sample covariance with the benchmark, not a lower mean. This is an ex-post regression diagnostic, not beta-neutral construction or a tradeable improvement.

Portfolio	N	Loading (MKT)	t (loading)	R2 (VTI market benchmark)	F
Combined model	213	-0.18	-4.21	0.205	17.71
BAB rebuilt	213	-0.18	-2.39	0.080	5.72
MOM rebuilt	213	-0.30	-2.89	0.105	8.35
QMJ rebuilt	213	-0.30	-3.77	0.160	14.18
VAL rebuilt	213	0.07	0.80	0.011	0.64
Standard model	230	-0.12	-2.46	0.070	6.03

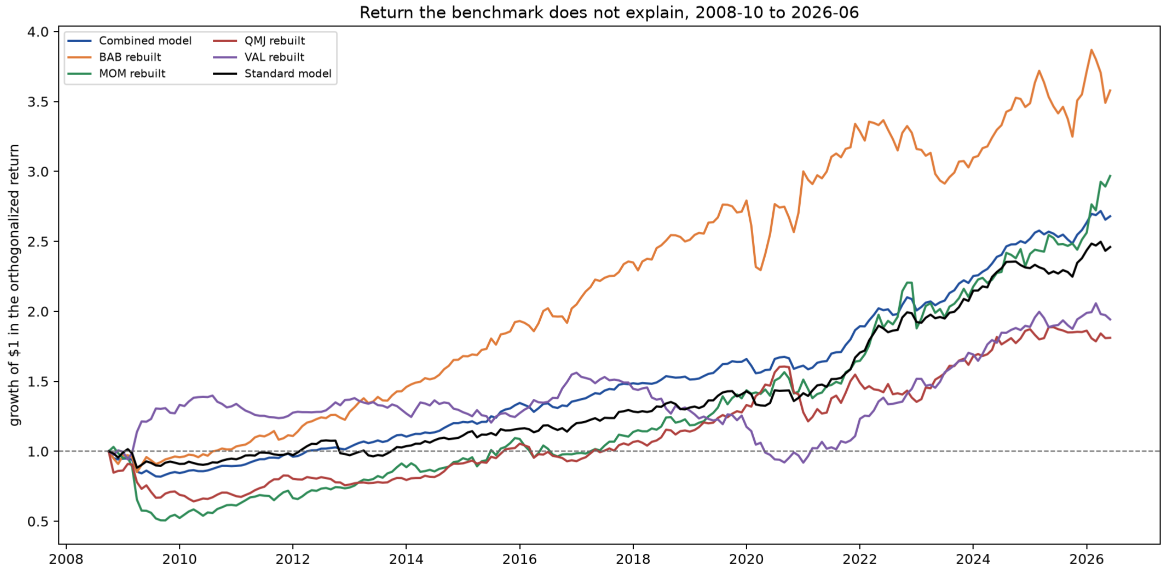

Figure 2: Returns orthogonalized to the VTI market benchmark: growth of one dollar in each book once the VTI market benchmark component is removed. A flat path earns nothing the benchmark does not already deliver.

The table summarizes the residual paths above.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	213	5.88%	5.88%	5.37%	1.09	4.24
BAB rebuilt	213	7.93%	7.74%	9.51%	0.83	4.09
MOM rebuilt	213	6.86%	6.09%	13.29%	0.52	1.91
QMJ rebuilt	213	4.21%	3.71%	10.49%	0.40	1.54
VAL rebuilt	213	4.50%	4.10%	9.80%	0.46	1.67
Standard model	230	5.21%	5.09%	6.88%	0.76	3.29

5 Attribution to configured factor controls

Regress each raw net return simultaneously on VAL+MOM+BAB+QMJ: $r_{p,t} = \alpha + \sum_k \beta_k f_{k,t} + \varepsilon_t$. The first table gives the loadings, with HAC t in brackets.

Portfolio	VAL	MOM	BAB	QMJ
Combined model	0.08 (2.45)	0.13 (4.17)	0.18 (5.77)	0.22 (6.95)
BAB rebuilt	0.02 (0.65)	0.08 (2.62)	0.71 (11.15)	0.10 (2.91)
MOM rebuilt	0.05 (0.51)	0.62 (5.86)	0.06 (0.65)	0.32 (3.47)
QMJ rebuilt	-0.15 (-3.08)	0.11 (2.46)	-0.02 (-0.31)	0.55 (6.93)
VAL rebuilt	0.41 (9.17)	-0.29 (-5.78)	-0.05 (-0.78)	-0.07 (-1.10)

Return contribution is $\beta_k \times 12 \times \overline{f_k}$. Explained plus alpha equals total annual return.

Portfolio	VAL	MOM	BAB	QMJ	Explained	Total
Combined model	-0.11%	0.74%	0.77%	1.28%	2.68%	3.16%
BAB rebuilt	-0.03%	0.43%	3.14%	0.56%	4.11%	5.04%
MOM rebuilt	-0.06%	3.53%	0.25%	1.84%	5.57%	2.79%
QMJ rebuilt	0.21%	0.64%	-0.09%	3.12%	3.88%	-0.40%
VAL rebuilt	-0.55%	-1.66%	-0.22%	-0.42%	-2.85%	5.21%

5.1 Simultaneous factor spanning and factor-orthogonalized returns

The same regression as a spanning test. R2 is the share of monthly net-return variance jointly explained by the factor controls.

Portfolio	N	R2 (4F)	F
Combined model	212	0.672	100.36
BAB rebuilt	212	0.827	241.69
MOM rebuilt	212	0.554	14.31
QMJ rebuilt	212	0.528	31.40
VAL rebuilt	212	0.579	34.23

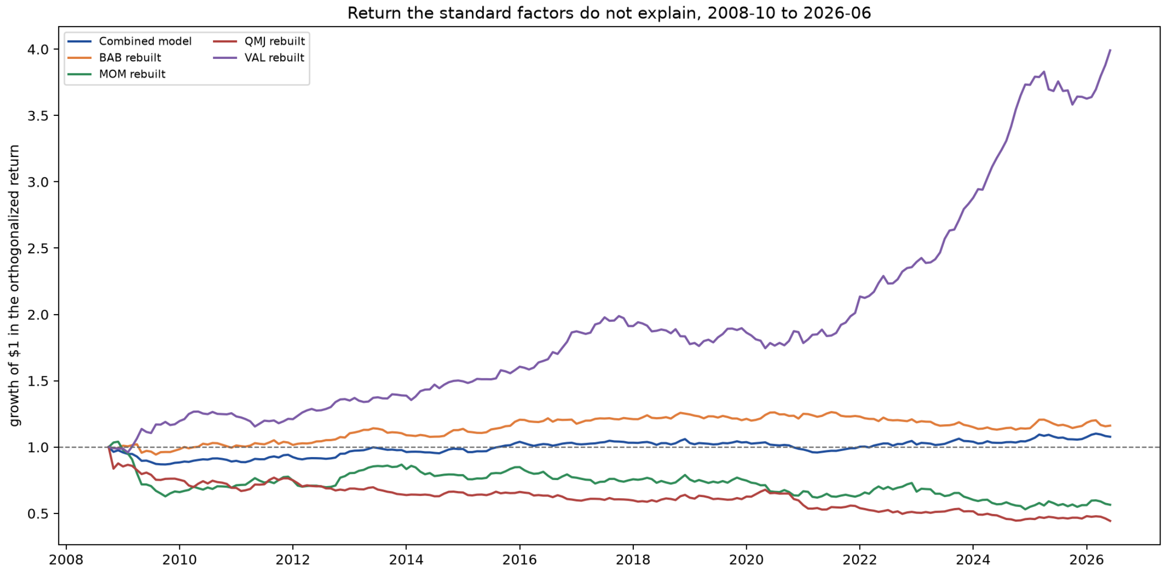

Figure 3: Returns orthogonalized to VAL+MOM+BAB+QMJ: growth of one dollar in each book once its factor exposures are removed. This is the Alpha column above, drawn as a path.

The table summarizes the factor-residual paths above.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	212	0.48%	0.43%	3.44%	0.14	0.51
BAB rebuilt	212	0.93%	0.85%	4.10%	0.23	1.01
MOM rebuilt	212	-2.78%	-3.18%	9.39%	-0.30	-1.10
QMJ rebuilt	212	-4.28%	-4.50%	7.83%	-0.55	-2.24
VAL rebuilt	212	8.06%	8.15%	6.37%	1.27	4.53

5.2 The new model against the reference model

Regress the combined model on Standard model as one return series. High R2 with zero alpha means the reference reproduces the model.

Portfolio	N	Alpha	t	R2 (Standard model)	F
Combined model	212	0.76%	0.65	0.631	96.79

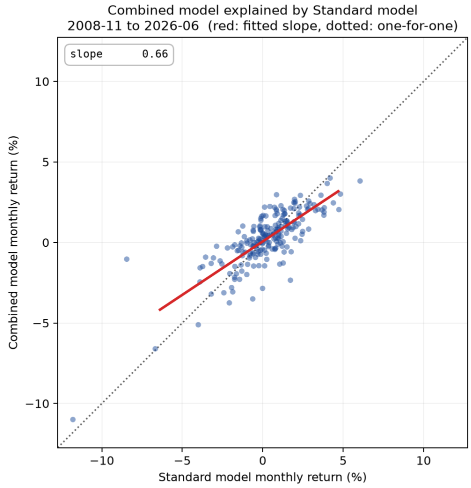

Figure 4: Each month of the combined model against Standard model, with the fitted slope. A tight cloud along the line is a model the reference reproduces; vertical spread is what it cannot.

5.3 Correlation and principal components

Check whether a sleeve duplicates a standard factor. The combined model is excluded because its overlap with its own sleeves is mechanical. PCA uses standardized sleeve and standard-factor returns over 212 common months. The share shown in the plot is the largest covariance-matrix eigenvalue divided by the sum of all eigenvalues: the fraction captured by one dominant co-movement, not return or regression R2. The combined model is excluded because its overlap with its own sleeves is mechanical.

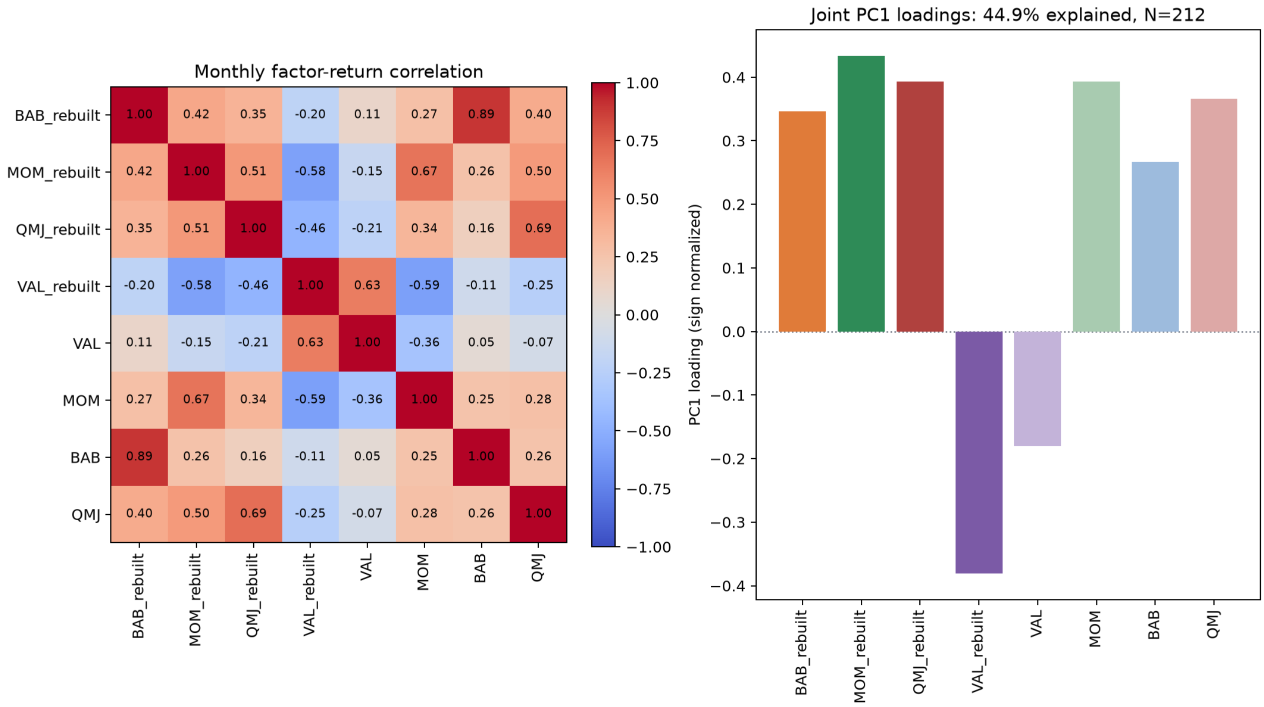

Figure 5: Monthly return correlations and sign-normalized first-principal-component loadings across the standalone sleeves and the supplied standard factors.

6 Signal persistence

Mean rank IC between formation exposure and a single later month's return. Flat profiles persist; fast decay implies faster rebalancing.

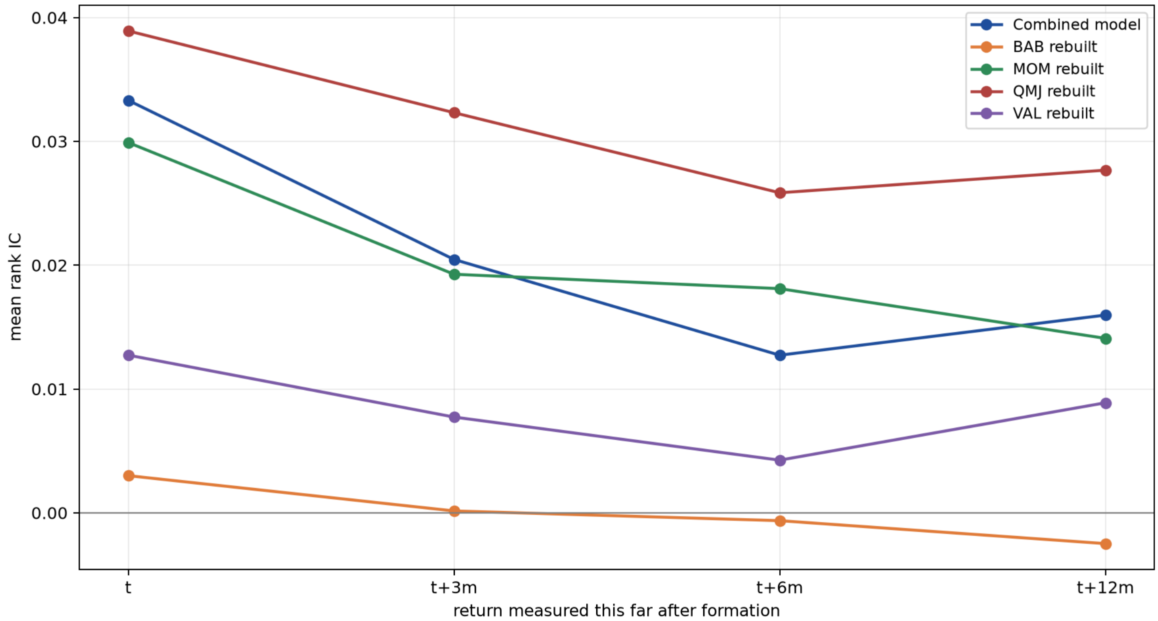

Figure 6: Mean rank information coefficient at each horizon. A line that stays flat is a signal that keeps working long after the month it is traded.

Portfolio	Horizon	Mean IC	t	Months
Combined model	t	0.0333	3.46	250
Combined model	t+3m	0.0205	2.15	247
Combined model	t+6m	0.0127	1.37	244
Combined model	t+12m	0.0160	2.09	238
BAB rebuilt	t	0.0030	0.30	250
BAB rebuilt	t+3m	0.0002	0.02	247
BAB rebuilt	t+6m	-0.0006	-0.06	244
BAB rebuilt	t+12m	-0.0025	-0.27	238
MOM rebuilt	t	0.0299	3.72	250
MOM rebuilt	t+3m	0.0193	2.33	247
MOM rebuilt	t+6m	0.0181	2.40	244
MOM rebuilt	t+12m	0.0141	2.06	238
QMJ rebuilt	t	0.0389	5.10	213
QMJ rebuilt	t+3m	0.0323	4.39	210
QMJ rebuilt	t+6m	0.0258	3.50	207
QMJ rebuilt	t+12m	0.0277	4.13	201
VAL rebuilt	t	0.0127	1.95	250
VAL rebuilt	t+3m	0.0077	1.13	247
VAL rebuilt	t+6m	0.0043	0.58	244
VAL rebuilt	t+12m	0.0089	1.27	238

7 Candidate comparisons and replacement

Standalone compares candidate with incumbent. In-model replaces the incumbent sleeve and rebuilds the standard model; diversification can reverse the standalone ranking.

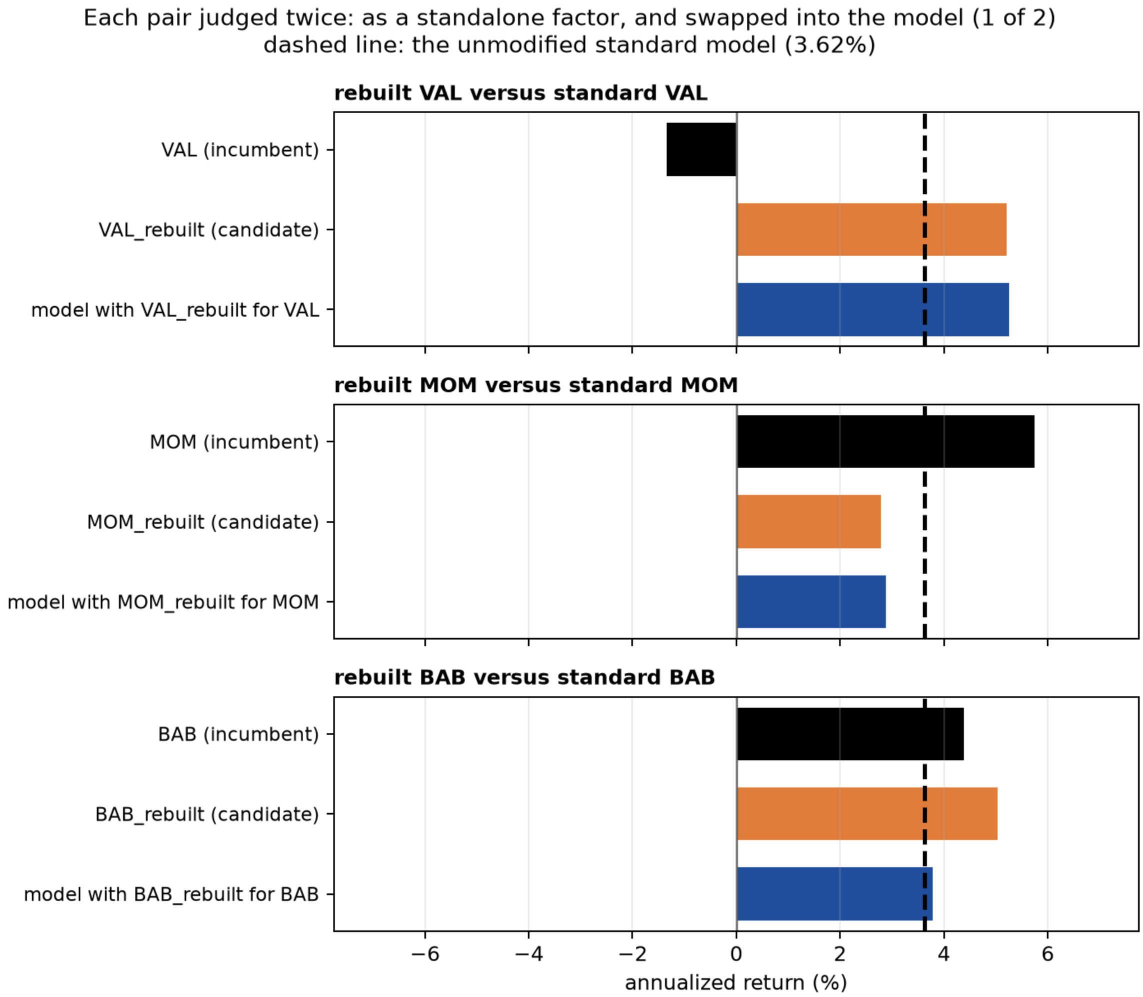

Figure 7: Each pair judged twice: the incumbent factor against the candidate standing alone, then the model with the candidate swapped in. The dashed line is the unmodified standard model.

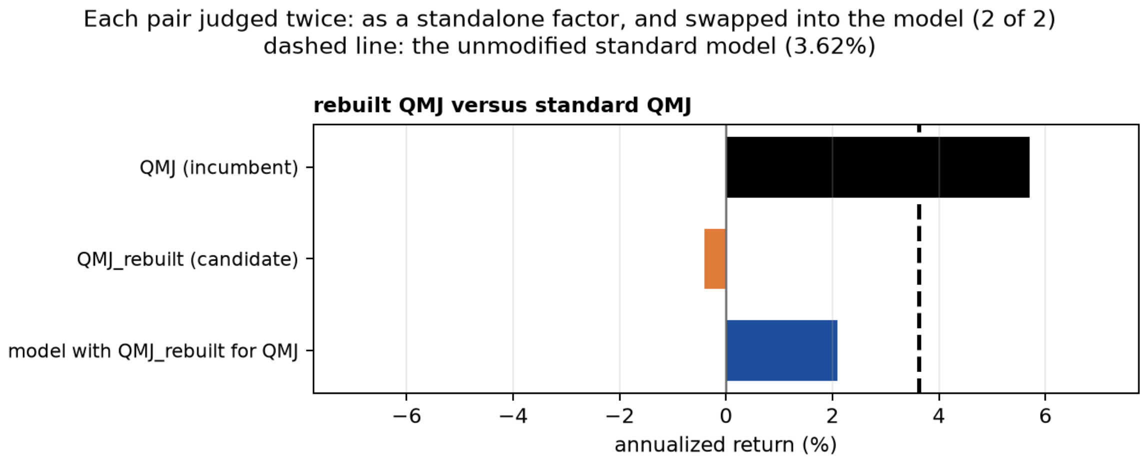

Figure 8: Candidate comparisons continued (2 of 2); axis and reference line are shared with the figure above.

Pair	Role	Portfolio	Ann. ret.	Change	Alpha	Change	t
rebuilt VAL versus standard VAL	incumbent	VAL	-1.35%	–	-1.31%	–	-0.34
	candidate	VAL rebuilt	5.21%	+6.56%	4.20%	+5.51%	1.45
	in model	Standard model, VAL rebuilt for VAL	5.26%	+1.64%	7.09%	+1.38%	4.99
rebuilt MOM versus standard MOM	incumbent	MOM	5.74%	–	8.08%	–	2.81
	candidate	MOM rebuilt	2.79%	-2.96%	7.13%	-0.95%	2.68
	in model	Standard model, MOM rebuilt for MOM	2.88%	-0.74%	5.47%	-0.24%	2.91
rebuilt BAB versus standard BAB	incumbent	BAB	4.39%	–	5.58%	–	1.88
	candidate	BAB rebuilt	5.04%	+0.65%	7.66%	+2.08%	3.25
	in model	Standard model, BAB rebuilt for BAB	3.79%	+0.16%	6.23%	+0.52%	4.06
rebuilt QMJ versus standard QMJ	incumbent	QMJ	5.71%	–	10.50%	–	3.76
	candidate	QMJ rebuilt	-0.40%	-6.11%	3.90%	-6.61%	1.48
	in model	Standard model, QMJ rebuilt for QMJ	2.10%	-1.53%	4.06%	-1.65%	2.90

8 Conclusion

Verdict. Not distinct: it clears the benchmark, but standard factors explain it. vs VTI market benchmark: alpha 5.88%, t=4.39; vs VAL+MOM+BAB+QMJ: alpha 0.48%, t=0.47.

9 Reproducibility

Run ID:

`20260826T141720Z_standard-model-published-construction_482bd6ce59`

Configuration hash: `482bd6ce59`. Gold version: `version=20260826T131322Z`. The run directory contains configuration, data products, figures, source, log, and PDF. Re-runs never overwrite it.